

IoT-Based Smart Waste Management System Using NodeMCU and Cloud Computing: A Case Study at PUST

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Abstract— This work aims to develop a centralized waste management system utilizing IoT and cloud computing to enhance waste collection efficiency and environmental sustainability. The system is initially implemented at Pabna University of Science and Technology (PUST), where smart dustbins are strategically installed in key locations, including the Administrative Building, Central Library, Sheikh Hasina Hall, Academic Buildings, Bangabandhu Sheikh Mujibur Rahman Hall, Central Cafeteria, Amphitheater, and Independence Square. Each dustbin is equipped with a pair of ultrasonic sensors, a gas sensor, a NodeMCU ESP8266 microcontroller, and a servo motor. These components enable real-time monitoring of waste levels and odor intensity, facilitating optimized waste collection. Unlike conventional systems, this project incorporates advanced features such as automated email alerts with precise location details for optimized garbage collection routes and a mobile application that enables users to locate the nearest available dustbin. The system utilizes cloud computing for data analysis, remote monitoring, and real-time updates, ensuring efficient waste disposal operations. By integrating IoT technology and predictive analytics, this system enhances cleanliness, hygiene, and waste management efficiency while promoting a sustainable campus environment.

Keywords— Smart dustbin, Waste management system, NodeMCU, IoT, Cloud Computing.

I. INTRODUCTION

Maintaining a clean and sustainable environment is largely dependent on efficient waste management, especially on university campuses where significant amounts of waste are produced on a daily basis [1-2]. The swift progress of technology has led to the emergence of inventive approaches that leverage cloud computing and IoT (Internet of Things) devices to optimize waste management procedures and foster ecological sustainability [3-5]. Pabna University of Science and Technology lacks a proper waste management system, leading to the indiscriminate disposal of garbage by students and staff, resulting in environmental degradation on campus. It becomes necessary to deal with this matter in order to

maintain a conducive learning environment, given that students at the university mainly place a premium on academics and research [6-7]. This project seeks to create an automatic waste monitoring system employing Internet of Things (IoT) technology. Waste levels in dustbins will be monitored realtime by system and data sent to cloud with the help of a Wi-Fi module [8]. Authorities will have access to this data, enabling them to select optimum waste collection vehicle scheduling and routing thereby conserving fuel and time. General public too can access this data through an app, for checking the status of dustbins around them and finding out the nearest empty dustbin - thus avoiding over-flowing ones. The system helps in efficient waste management, while also increasing awareness about environment [9].

Our main contributions in this work can be outlined as follows:

- **Development of an IoT-Enabled Smart Waste Management System:** Designed and implemented an IoT-based waste monitoring system using ultrasonic sensors, a gas sensor, a NodeMCU ESP8266 microcontroller, and a servo motor for automated waste level detection and odor monitoring.
- **Integration of Cloud Computing for Real-Time Data Processing:** Established a cloud-based platform for real-time data acquisition, storage, and analysis, enabling remote monitoring of dustbin status and waste levels.
- **Implementation of an Intelligent Waste Collection Strategy:** Developed an automated notification system that sends email alerts with precise location details when a dustbin reaches its threshold capacity, ensuring timely waste disposal and optimized collection routes.
- **Development of a User-Friendly Mobile Application:** Designed an Android-based mobile application to help users locate the nearest empty dustbin, promoting efficient waste disposal and improving overall campus hygiene.

- **Enhancing Sustainability and Hygiene on Campus:** Implemented a smart waste management system at Pabna University of Science and Technology (PUST) to improve cleanliness, optimize resource utilization, and contribute to environmental sustainability.
- **Utilization of Predictive Analytics for Waste Management Optimization:** Leveraged cloud-based analytics to identify waste generation patterns, optimize collection schedules, and enhance decision-making for future smart waste management solutions.

These contributions demonstrate the effectiveness of integrating IoT and cloud computing in waste management, providing a scalable and efficient solution for smart cities and university campuses.

The rest of the paper is organized as follows: Section II discusses related works, Section III provides a general description of the key components of the prototype, and Section IV details the proposed system, including the workflow diagram. Section V presents the implemented prototype and real-time testing, while Section VI concludes the paper.

II. LITERATURE REVIEW

Intelligent waste management systems have been widely explored to enhance waste collection efficiency and environmental sustainability. In [10], an Arduino Ethernet Shield was used to collect and transmit garbage level data from dustbins to the Ubidots IoT Cloud, where the bin's fill status was visualized on a dashboard, and alert notifications were sent to garbage collectors. Similarly, in [11], a smart garbage management system employed Zigbee and GSM technologies, with sensors detecting waste levels and an ARM7 processor triggering GSM notifications. Zigbee facilitated short-range communication, while GSM extended coverage, and image analysis optimized collection routes to reduce costs and improve cleanliness. In [12], an ultrasonic detector and a NodeMCU were implemented to monitor garbage levels in dustbins, with real-time data uploaded to a Firebase database and displayed on an Android application and user interface. Meanwhile, [13] introduced a system integrating cameras, RFID, GPS, and GIS for waste monitoring. RFID readers mounted on collection trucks retrieved bin data, while a control unit processed information via SMS. GPS and GPRS-based mapping enabled location analysis, with a centralized workstation compiling records and tracking dustbins and collection trucks. Further advancements were seen in [14], where ultrasonic detectors, an Arduino, and GPS modules relayed bin status to sanitation workers, ensuring timely waste collection. An IR sensor and a DC motor were also incorporated to automate bin lid operation when users approached. In [15], an Arduino microcontroller, ultrasonic detectors, a GSM module, and an LCD screen were utilized to monitor and display garbage levels. The data was transmitted to waste collectors via GSM technology; however, the high cost and power inefficiency of GSM modules limit their large-scale deployment in metropolitan areas.

Despite these advancements, existing studies have not addressed the simultaneous optimization of waste collection routes based on both garbage accumulation and odor levels. Additionally, no system has been proposed to assist users in locating the nearest available empty dustbin for convenient waste disposal.

III. KEY COMPONENTS

This section provides a brief overview of essential components utilized in the system, including the Ultrasonic Sensor, Gas Sensor, NodeMCU, and Servomotor. These components play a crucial role in detecting waste levels, monitoring gas emissions, enabling wireless communication, and automating bin operations, respectively.

A. Ultrasonic Sensor(HC-SR04)

As shown in Fig. 1, the ultrasonic sensor is a widely used distance-measuring module that operates at 5V DC with a frequency of 40 kHz and a detection range of 2 cm to 400 cm. It consists of four pins: V_{CC}, Trigger, Echo, and Ground. The sensor calculates the distance to an object by measuring the time interval between transmitting ultrasonic waves and receiving the reflected signal using the Echo pin. The distance is determined using the formula: $L = 1/2 \times t \times c$ where c is the speed of sound (343 m/s at 20°C) and t is the time taken for the wave to return. The ultrasonic sensor is widely employed in various applications, including robotics, parking systems, security, automation, and medical devices. The HC-SR04 model is particularly compatible with Arduino, Raspberry Pi, and NodeMCU, making it suitable for contactless distance measurement in numerous applications.



Fig. 1. HC-SR04 Ultrasonic Detector

B. MQ-2 Gas Sensor

As shown in Fig. 2, the MQ-2 gas sensor module is used to detect combustible gases such as LPG, methane, butane, hydrogen, carbon monoxide (CO), alcohol, and smoke within a detection range of 300 to 10,000 ppm. It operates at 5V DC and provides both analog and digital outputs. The MQ-2 sensor functions by measuring the resistance changes in its metal oxide semiconductor (SnO₂) layer when exposed to gases. This change in resistance generates a proportional voltage output corresponding to the gas concentration. The MQ-2 sensor is widely used in applications such as gas leak detectors, fire alarms, air quality monitoring, industrial safety systems, and breath analyzers. It is highly sensitive, offers a fast response time, and can be easily interfaced with Arduino, Raspberry Pi, and other microcontrollers, making it an effective solution for safety and environmental monitoring applications.



Fig. 2. MQ-2 Gas Sensor

C. NodeMCU ESP8266

As shown in Fig. 3, the NodeMCU ESP8266 is a versatile development board based on the ESP8266 Wi-Fi module, known for its affordability and robust connectivity features. It integrates a microcontroller unit (MCU) with built-in Wi-Fi, enabling seamless internet connectivity, making it well-suited for Internet of Things (IoT) applications. NodeMCU is widely preferred for its ease of use and compatibility with the Arduino IDE and Lua scripting language, allowing developers to create a diverse range of applications, including home automation systems, sensor networks, and smart devices. Its versatility makes it a popular choice for both prototyping and deployment in various IoT-based projects.

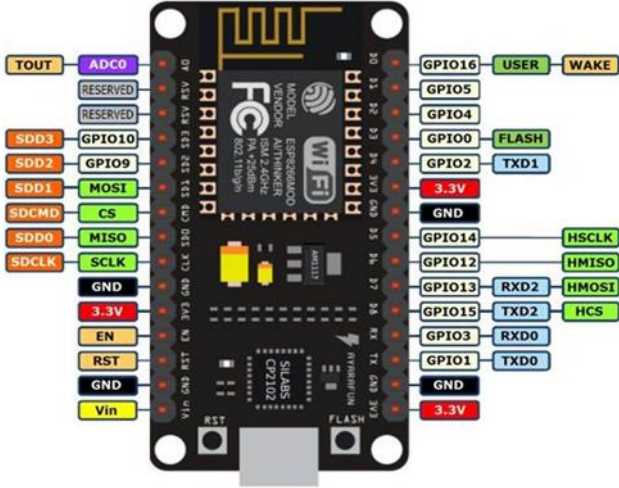


Fig. 3. NodeMCU ESP8266

D. SG90 Servo Motor

As shown in Fig. 4, the SG90 servo motor is a widely used component in robotics and hobbyist projects. It features three pins: signal, power, and ground. The signal pin receives control pulses to determine the motor's position, while the power pin connects to a power supply, typically ranging from 4.8V to 6V. The ground pin completes the circuit by connecting to the negative terminal of the power source. By adjusting the width of the control pulses, users can precisely control the servo motor's position, making it ideal for applications requiring controlled movement, such as robotic arms, remote-controlled vehicles, and automation systems.



Fig. 4. SG90 Servo Motor

IV. PROPOSED IOT-BASED SMART WASTE MANAGEMENT SYSTEM

The proposed waste management system integrates IoT and Cloud computing technologies for real-time monitoring of waste levels in dustbins and determining the most efficient garbage collection routes. Each dustbin is equipped with an ultrasonic sensor, a gas sensor, a servo motor, and a NodeMCU Wi-Fi module. The waste level data is uploaded to

Cloud storage and displayed on a dashboard accessible by authorities through an Android application. Fig. 5 illustrates the conceptual diagram of the proposed smart waste management system for Pabna University of Science and Technology (PUST), while Fig. 6 presents its block diagram. These figures depict the system's key components and their interactions, ensuring efficient waste monitoring and management.

1. NodeMCU ESP8266: Central to the system, this microcontroller communicates with various sensors and cloud platforms.
2. Ultrasonic Sensor 1: Measures the fill level of the waste bin by detecting the distance between the waste and the sensor.
3. Ultrasonic Sensor 2: Detects the presence of nearby objects, such as when someone approaches the dustbin.
4. MQ-2 Gas Sensor: Monitors the air quality by detecting gases emitted from the waste, particularly harmful gases that may indicate potential hazards.
5. Servo Motor: Activates the dustbin lid, allowing for automatic opening and closing.

The flow of operations of the proposed smart waste management system is as follows:

1. Data Collection:

- Ultrasonic Sensor 1 measures the distance from the top of the dustbin to the waste to determine the fill level.
- Ultrasonic Sensor 2 detects the presence of objects or people near the dustbin.
- MQ-2 Gas Sensor continuously monitors the air quality by detecting gases from the waste (e.g., methane or carbon dioxide).

2. Data Processing:

- The NodeMCU ESP8266 collects sensor data, such as distance, presence, and gas concentration, and processes it locally.
- Based on predefined thresholds (e.g., a certain fill level or high gas concentration), NodeMCU determines whether intervention is needed (e.g., triggering a collection process or alerting authorities).

3. Action Triggering:

If the fill level reaches the threshold, or if gas levels exceed safe limits, NodeMCU initiates appropriate actions, such as:

- Sending alerts (via email or notifications) to waste management authorities or service providers (e.g., waste truck drivers) using the Blynk server.
- Activating alarms or triggering the servo motor to open the dustbin lid automatically.

4. Cloud Communication:

NodeMCU transmits real-time sensor data to a cloud platform (e.g., Blynk) via Wi-Fi.

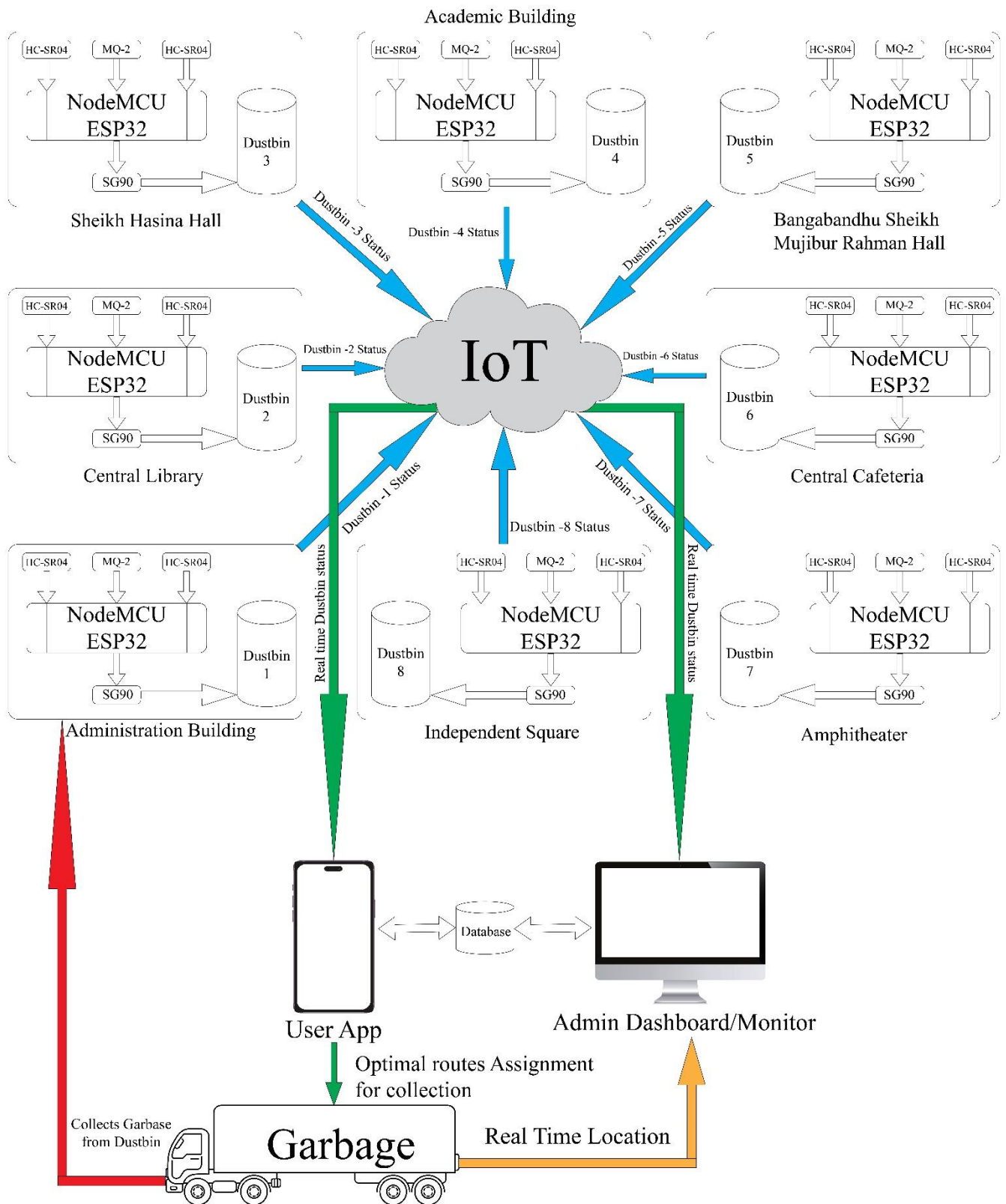


Fig.5. Conceptual Architecture of the proposed IoT-based smart waste management system.

- Waste management authorities or service providers can remotely monitor the status of dustbins via a web or Android application. This includes viewing fill levels, gas concentrations, and receiving real-time alerts for timely intervention.

5. Data Analytics and Optimization:

- The cloud platform stores and analyzes historical data, offering insights into waste generation patterns and trends.

- Predictive analytics can forecast future waste generation, helping optimize resource allocation, collection schedules, and routes.
- Data-driven insights continuously improve waste management strategies and infrastructure through feedback loops.

Fig. 7 illustrates the flowchart of how live sensor data is sent to the user interface through the Blynk platform. Blynk is an open-source IoT platform that allows for the creation of custom Android apps, enabling users to visualize and interact with sensor data in real-time. Users can view various parameters, such as distance, gas level, and the servo motor's position, directly on their mobile devices. Fig. 8 demonstrates the visual representation of the Blynk app interface, which provides users with a graphical display of real-time sensor data. It shows the distance and dust level measured by the ultrasonic sensor, gas levels detected by the MQ-2 sensor, and the servo motor's angle (for dustbin lid operation). This interface empowers users to efficiently manage and monitor waste levels.

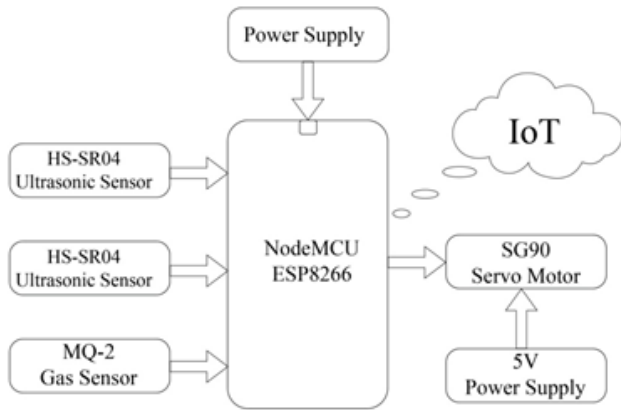


Fig. 6. Block Diagram of the proposed system.

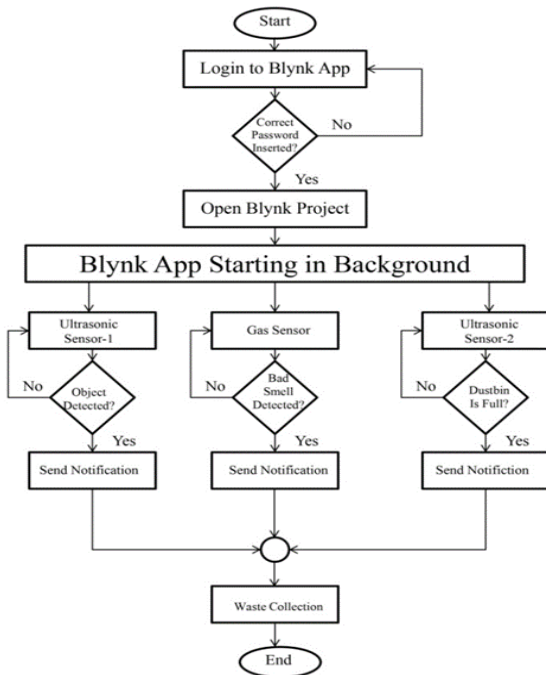


Fig. 7: Flow Chart of the proposed system

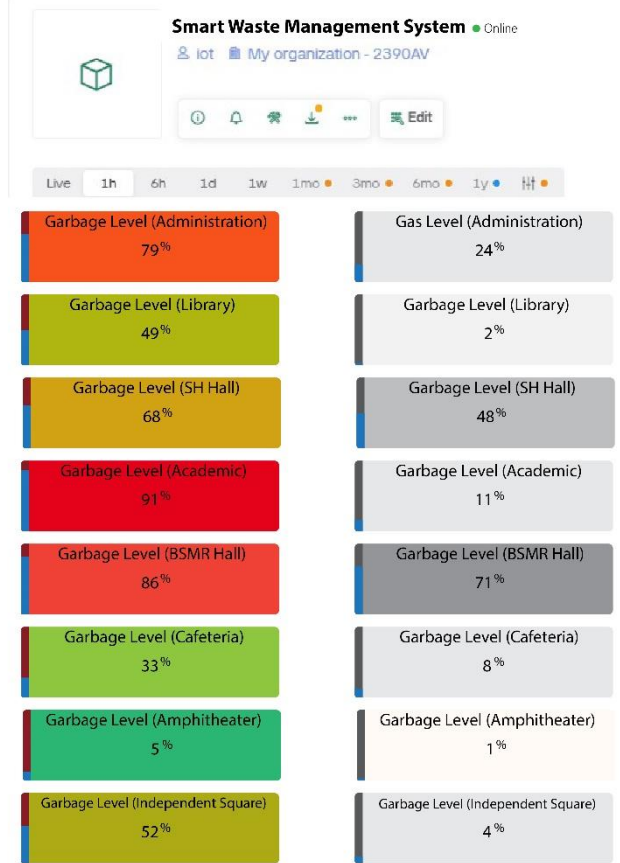
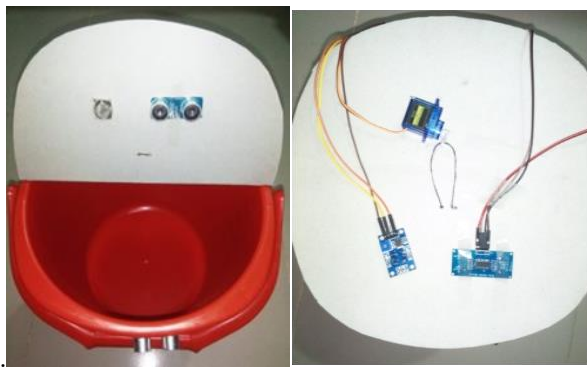


Fig. 8. Real-time status of the dustbin displayed in Blynk Web Dashboard.

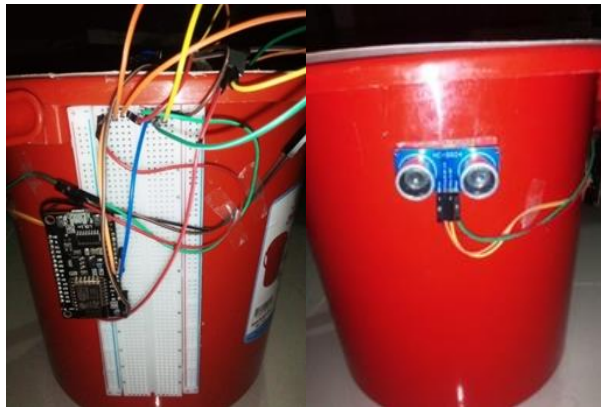
V. PROTOTYPE DESIGN AND VALIDATION

A working prototype of the smart dustbin was developed and tested. The internal structure of the model is shown in Fig. 9(a), displaying the placement of the ultrasonic and gas detectors at the top of the waste compartment. Fig. 9(b) provides a top view of the prototype, highlighting the ultrasonic and gas sensors, as well as the servo motor used for lid operation. The back view of the prototype is presented in Fig. 9(c), showing the mounting of the NodeMCU board, ultrasonic sensors, gas sensor, and servo motor on the dustbin. The system's functionality was validated using the Arduino console, which printed real-time sensor data to confirm the accuracy of measurements. These sensor data were transmitted to the Blynk platform for visualization on both mobile applications and web pages. A printout of the Arduino console data is presented in Fig. 9(d), showing the respective sensor readings and confirming the system's effectiveness for intelligent garbage monitoring. The system was evaluated in a real environment and validated, demonstrating its readiness for integration into a smart dustbin monitoring solution for PUST



(a) Inner view

(b) Top view.



(c) Back view

(d) Front view

Fig. 13. Prototype of IoT-based smart dustbin

VI. CONCLUSION

This smart waste management system based on NodeMCU and cloud computing has been specifically designed for Pabna University of Science and Technology (PUST). The main objective for implementing this system at PUST is to increase the efficiency of waste management on the campus so that students and staff can have a cleaner and more organized environment. Generally, the traditional method of waste management at the university faces several challenges like overflowing bins, delayed pickups, and inefficient waste collection. The IoT-based smart dustbin will continuously monitor the waste level in real-time, so the system will ensure fewer pickups of the bins only when required, thereby curtailing waste overflow and unnecessary collection trips. The future plan for the system aims to integrate it into the Blynk platform, thus enabling the staff and management of the university to track the fill levels of dustbins in real-time from a mobile app and website dashboard. Additionally, the system will gather data that will be of great help for the waste management team to set optimized collection schedules for better management of waste on campus. This advancement therefore aims to provide a more sustainable, efficient, and cleaner campus ambiance to current and future students.

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